

2.1 System software v/s Application software

A software is a set of instructions or programs instructing a computer to do specific tasks. Software is basically a generic term used to describe computer programs. In general scripts, applications, programs and a set of instructions are terms often used to describe a software.

On the basis of language in which software is developed and platform which is required for its execution, we can classify software into two divisions: System software and Application software. The following are some basic differences between a System software and an Application software.

Key	System Software	Application Software
Definition	System software is an interface between application software and system.	Application software is the type of software which runs as per user request. It runs on a platform which is provided by the system software.
Development Language	In general, system software are developed in low level language which is more compatible with the system hardware in order to interact with.	While in case of Application software high level language is used for their development as they are developed as some specific purpose software.
Usage	System software is used for operating computer hardware.	Application software is used by user to perform specific tasks.
Installation	System software are installed on the computer when operating system is installed.	Application software are installed according to user's requirements.
User interaction	System software are specific to system hardware. Therefore, very less or no user interaction available in system software.	Application software user can interact with it using a user interface.
Dependency	System software can run independently. It provides platform for running application software.	Application software can't run independently. They can't run without the presence of a system software.
Examples	Some examples of system software's are compiler, assembler, debugger, driver, etc.	On other hand some examples of application software's are word processor, web browser, media player, etc.

2.2 Operating System Abstractions

An Operating System (OS) is an interface between a computer user and computer hardware. An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers. Some popular Operating Systems include Linux Operating System, Windows Operating System, VMS etc.

Definition

An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.

2.2.1 Functions of an Operating System

An operating system performs various functions as shown in Figure 2.1. The following are some of important functions of an operating System.

- Memory Management
- Processor Management
- Device Management
- File Management
- Security
- Control over system performance
- Job accounting
- Error detecting aids
- Coordination between other software and users



Fig. 2.1 Functions of an Operating System

Memory Management

Memory management refers to management of Primary Memory or Main Memory. Main memory is a large array of words (set of 2 bytes) or bytes where each word or byte has its own address. Main memory provides a fast storage that can be accessed directly by the CPU. For a program to be executed, it must be in the main memory. An Operating System does the following activities for memory management:

- Keeps tracks of primary memory, i.e., which parts of it are in use by whom, which parts are not in use.
- In multiprogramming, the OS decides which process will get memory when and how much.
- Allocates the memory when a process requests it to do so.

- De-allocates the memory when a process no longer needs it or has been terminated.

Processor Management

In multiprogramming environment, the OS decides which process gets the processor when and for how much time. This function is called **process scheduling**. An Operating System does the following activities for processor management:

- Keeps tracks of processor and status of processes. The program responsible for this task is known as **traffic controller**.
- Allocates the processor (CPU) to a process.
- De-allocates processor when a process is no longer required.

Device Management

An Operating System manages device communication using respective drivers. It does the following activities for device management:

- Keeps tracks of all devices. Program responsible for this task is known as the I/O controller.
- Decides which process gets the device when and for how much time.
- Allocates the device in the efficient way.
- De-allocates devices.

File Management

A file system is normally organized into directories for easy navigation and usage. These directories may contain files and other directions. An Operating System does the following activities for file management:

- Keeps track of information, location, uses and status etc.
- Decides who gets the resources.

- Allocates the resources.
- De-allocates the resources.

Other Important Activities

Following are some of the important activities that an Operating System performs:

- **Security:** By means of password and similar other techniques, it prevents unauthorized access to programs and data.
- **Control over system performance:** Recording delays between request for a service and response from the system.
- **Job accounting:** Keeping track of time and resources used by various jobs and users.
- **Error detecting aids:** Production of dumps, traces, error messages, and other debugging and error detecting aids.
- **Coordination between other software and users:** Coordination and assignment of compilers, interpreters, assemblers and other software to the various users of the computer systems.

2.2.2 Examples of Operating Systems

Linux and Windows both Operating systems have their advantages and disadvantages. Windows is simple to use but is not a free and open-source OS, whereas Linux is free, open source, customizable and secure but kind of complex for the users having no programming background. Linux is more reliable than windows. The prior difference between Linux and Windows operating system is that Linux is totally free of cost whereas windows is marketable operating system and is costly. An operating system is a program intended to control the computer hardware and behave as an intermediary between user and hardware.

2.3 Firmware

In electronic systems and computing, **firmware** is a tangible electronic component with embedded software instructions, such as a BIOS. Typically, those software instructions are used to tell an electronic device how to operate. Typical examples of devices containing firmware are embedded systems (such as traffic lights, consumer appliances, and digital watches), computers, computer peripherals, mobile phones, and digital cameras. The firmware contained in these devices provides the control program for the device.

Firmware is stored in non-volatile memory devices such as ROM, EPROM, or flash memory. Changing the firmware of a device may rarely or never be done during its economic lifetime; some firmware memory devices are permanently installed and cannot be changed after manufacture. Common reasons for updating firmware include fixing bugs or adding features to the device. This may require ROM integrated circuits to be physically replaced, or flash memory to be reprogrammed through a special procedure. Firmware such as the ROM BIOS of a personal computer may contain only elementary basic functions of a device and may only provide services to higher-level software. Firmware such as the program of an embedded system may be the only program that will run on the system and provide all of its functions.

Interesting Information

BIOS, computing, stands for Basic Input/ Output System. The BIOS is a computer program embedded on a chip on a computer's motherboard that recognizes and controls various devices that make up the computer.

Do you know?

The first operating system was created by General Motors in 1956 to run a single IBM central computer. In the 1960s, IBM was the first computer manufacturer to take on the task of developing operating systems and began distributing operating systems included in its computers.

2.4 Installing Ubuntu

The Ubuntu desktop is easy to use, easy to install and includes everything you need to run your organization, school, home or enterprise. It's also open source, secure, accessible and free to download.

It is easy to install Ubuntu from a DVD. Follow the following steps to install Ubuntu:

1. Put the Ubuntu DVD into your optical/DVD drive.
2. Restart your computer.

As soon as your computer boots, you'll see the welcome window (Figure 2.2).

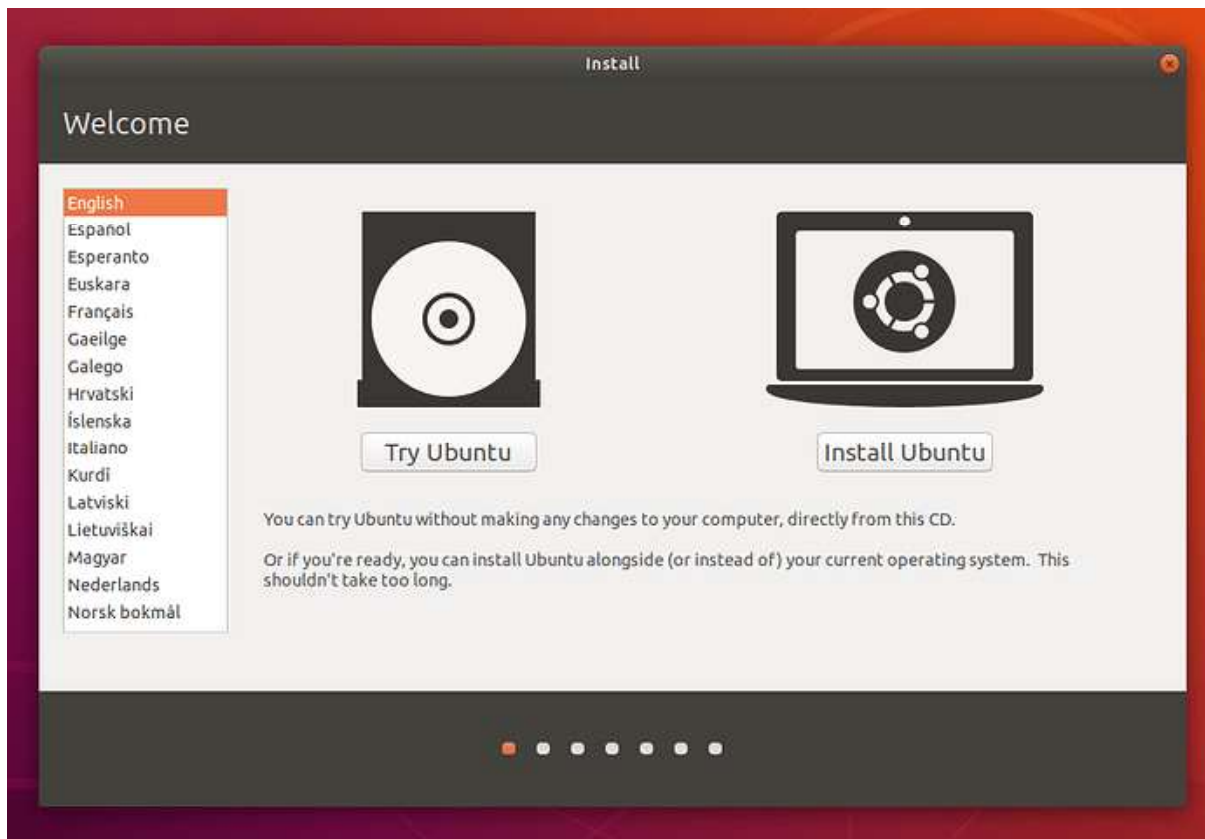


Fig. 2.2 Ubuntu Installation: Welcome Window

Select 'Install Ubuntu', and press return to launch the desktop installer automatically. Alternatively, select the first option, 'Try Ubuntu without installing', to test Ubuntu.

Boot from USB Flash Drive

Most computer sbboot from USB automatically. Simply insert the USB flash drive and turn on your computer. You should see the same welcome window we saw in the previous ‘Install from DVD’ step, prompting you to choose your language and install or try the Ubuntu desktop. If your computer doesn’t automatically boot from USB, try holding F12 when your computer starts. This will allow you to select the USB device from a system-specific boot menu. F12 is the most common key for bringing up your system’s boot menu, but Escape+F2 and F10 are common alternatives.

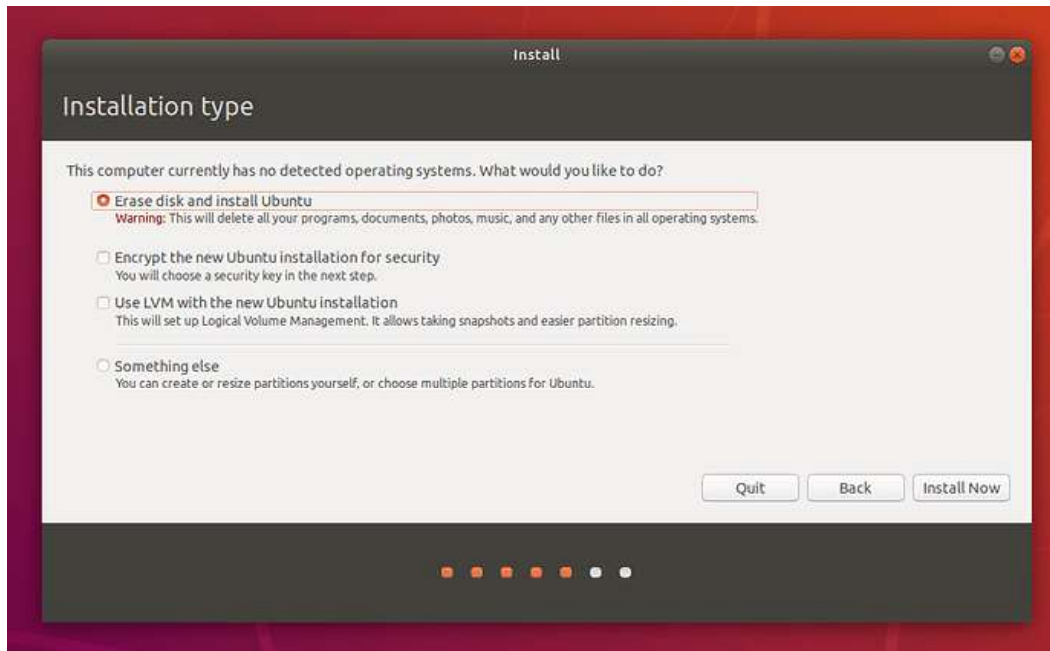
Prepare to install Ubuntu

After booting from selected storage media, installation wizard asks you to select your keyboard layout. If the installer doesn’t guess the default layout correctly, use the ‘Detect Keyboard Layout’ button to run through a brief configuration procedure. After selecting ‘Continue’ you will be asked ‘*What apps would you like to install to start with?*’ The two options are ‘Normal installation’ and ‘Minimal installation’. The first is the equivalent to the old default bundle of utilities, applications, games and media players. The second takes considerably less storage space and allows you to install only the required components.

Below the “installation-type” question are two checkboxes; one to enable updates while installing and another to enable third-party software.

Allocate Drive Space

Use the checkboxes to choose whether you’d like to install Ubuntu alongside another operating system, delete your existing operating system and replace it with Ubuntu, or — if you’re an advanced user — choose the ‘Something else’ option (Figure 2.3).



Do you know?

The word "Ubuntu" is an ancient Zulu and Xhosa word that means "humanity to others". Ubuntu also means "I am what I am because of who we all are". It was chosen because these sentiments precisely describe the spirit of the Ubuntu distribution.

Begin Installation

After configuring storage, click on the '*Install Now*' button. A small pane will appear with an overview of the storage options you've chosen, with the chance to go back if the details are incorrect. Click Continue to apply those changes in place and start the installation process.

Select Location

If you are connected to the internet, your location will be detected automatically. Check your location is correct and click 'Forward' to proceed. If you're unsure of your time zone, type the name of a local town or city or use the map to select your location.

Login Details

In the next step, a login window (Figure 2.4) will appear. Enter your name and the installer will automatically suggest a computer name and username. Username will be

your login and account name. Next, enter a strong password. You can also choose to enable automatic login and home folder encryption.

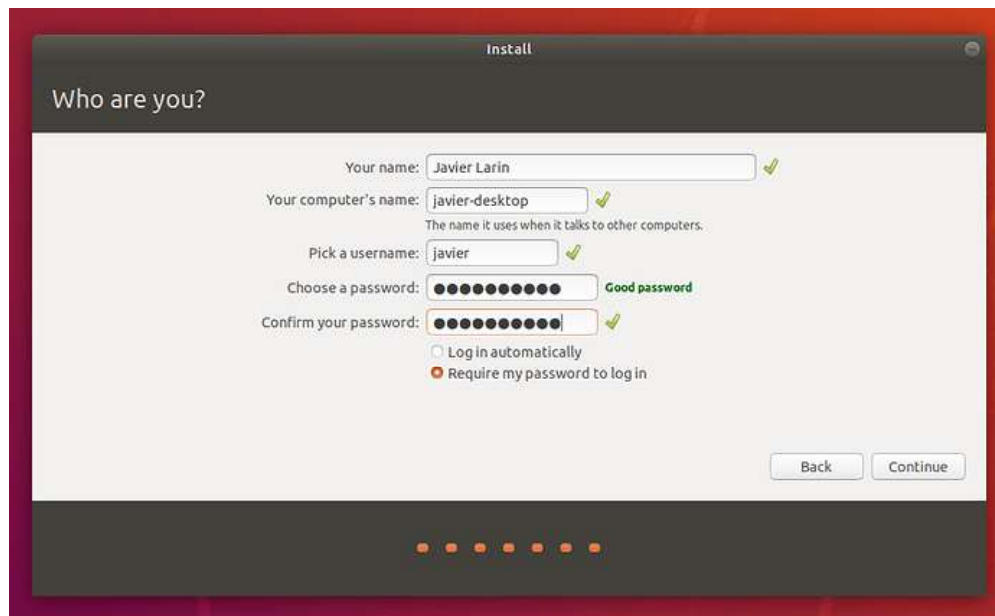


Fig. 2.4 Ubuntu Installation: Login Details

After everything has been installed and configured, a small window (Figure 2.5) will appear asking you to restart your machine. Click on Restart Now and remove either the DVD or USB flash drive when prompted. If you initiated the installation while testing the desktop, you also get the option to continue testing.

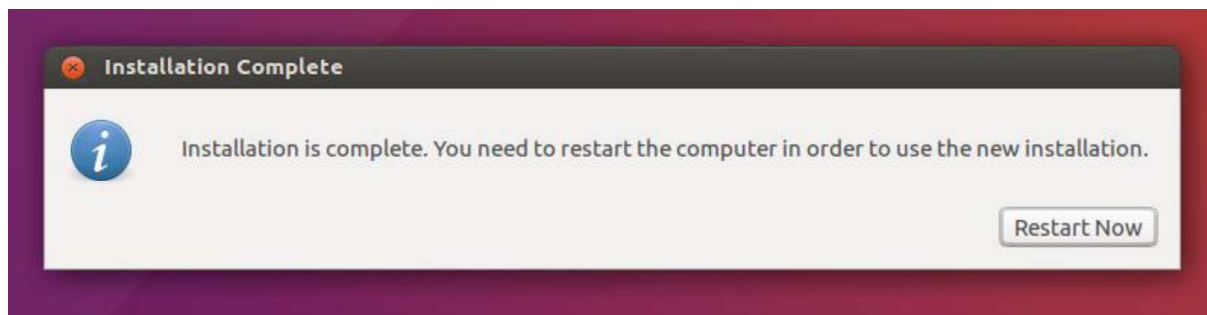


Fig. 2.4 Ubuntu Installation: Restart Window

2.5 Installing Windows

Create USB installation media: Visit Microsoft's Windows 10 download page and select "Download tool now" under the "create Windows 10 installation media" section. Transfer the downloaded installer tool to a USB drive.

Run the installer tool: Open the installer tool by clicking on it. Accept Microsoft's terms, and then select "Create installation media for another PC" on the "What do you want to do?" page. After selecting which language you want, and which edition you want as well (32-bit or 62-bit), you'll be asked what type of media you want to use.

Installing from a USB drive is definitely the preferred option but you can also install from a CD or ISO file. Once you choose your device, the installer tool will download the required files and put them onto your drive.

Use your installation media: Insert your installation media into your device and then access the computer's **BIOS or UEFI**. These are the systems that allow you to control your computer's core hardware. The process of accessing these systems is unique to each device, but the manufacturer's website should be able to give you a helping hand here. Generally, you'll need to press the F2, F12 or Delete keys as your computer boots up.

Change your computer's boot order: Once you have access to your computer's BIOS/UEFI you'll need to locate the settings for boot order. You need the Windows 10 installation tool to be higher up on the list than the device's current boot drive: this is the SSD or HDD that your existing OS is stored on. You should move the drive with the installer files to the very top of the boot order menu. Now, when you restart your device the Windows 10 installer should load up first.

Restart your device: Save your settings in the BIOS/UEFI and reboot your device.

Complete the installation: Your device should now load up the Windows 10 installation tool on restart. This will guide you through the rest of the installation process.

Wait for your computer to restart. Once your computer finishes restarting, you'll see a window here with your geographical data. You're now ready to begin setting up your Windows 10 installation. Follow the following steps to complete the installation of Windows 10.

Click Install Now. It's in the middle of the window (Figure 2.5).

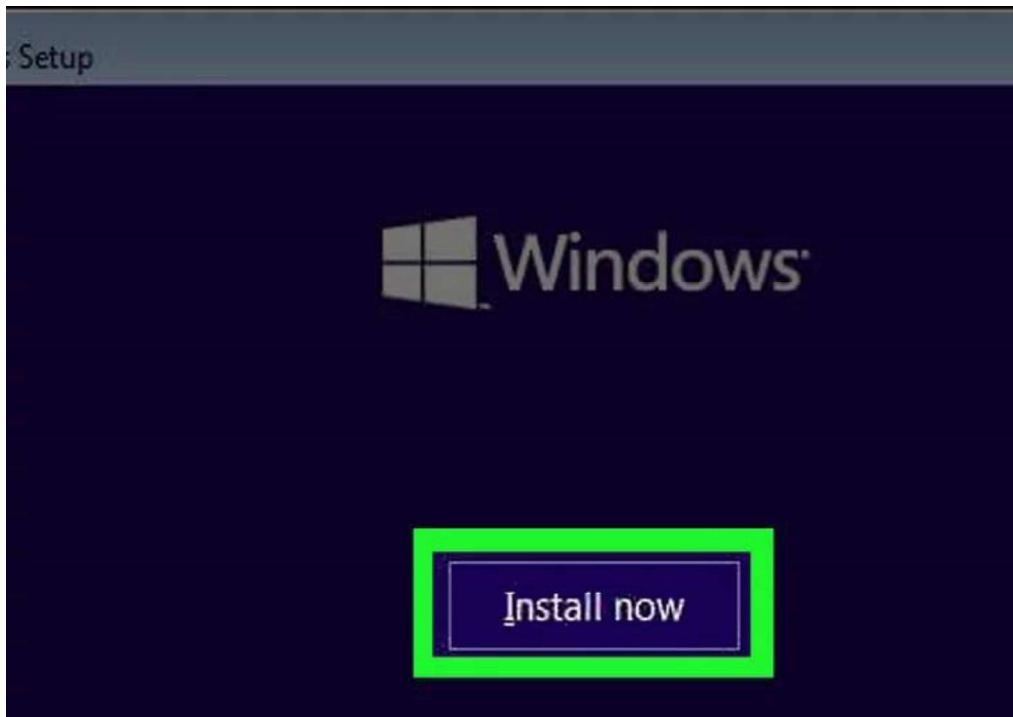


Fig. 2.5 Windows Installation: Install Now

In the next window (Figure 2.6), enter your Windows 10 key, then click Next. If you don't have a Windows 10 key, instead click Skip in the bottom-right corner of the screen.

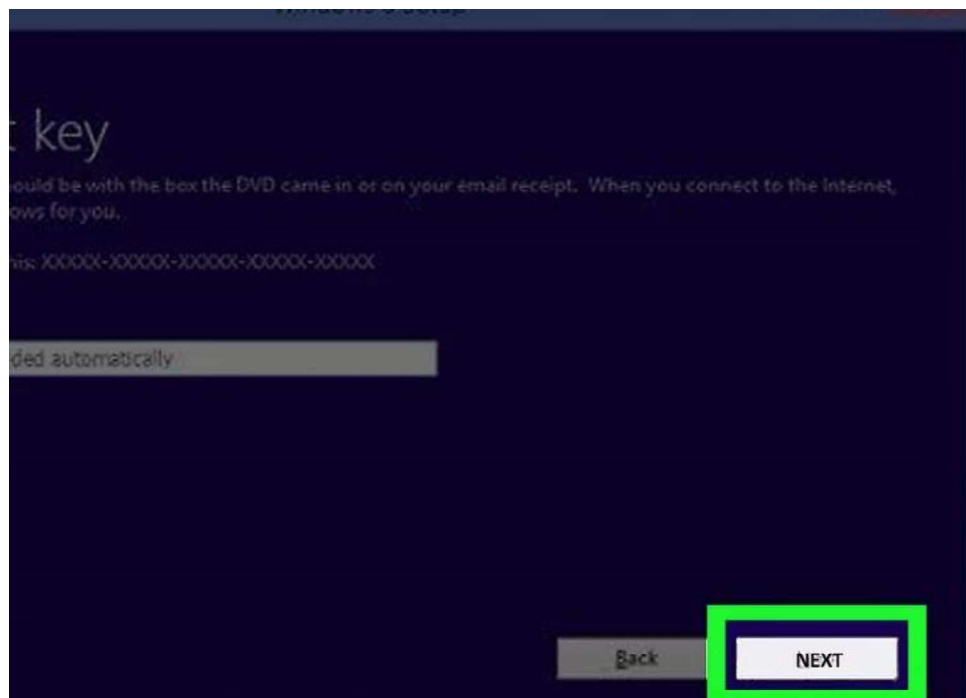


Fig. 2.6 Windows Installation: Enter Key

In the next window (Figure 2.7), click the "Accept" box and click Next. This will indicate that you accept the terms of use.

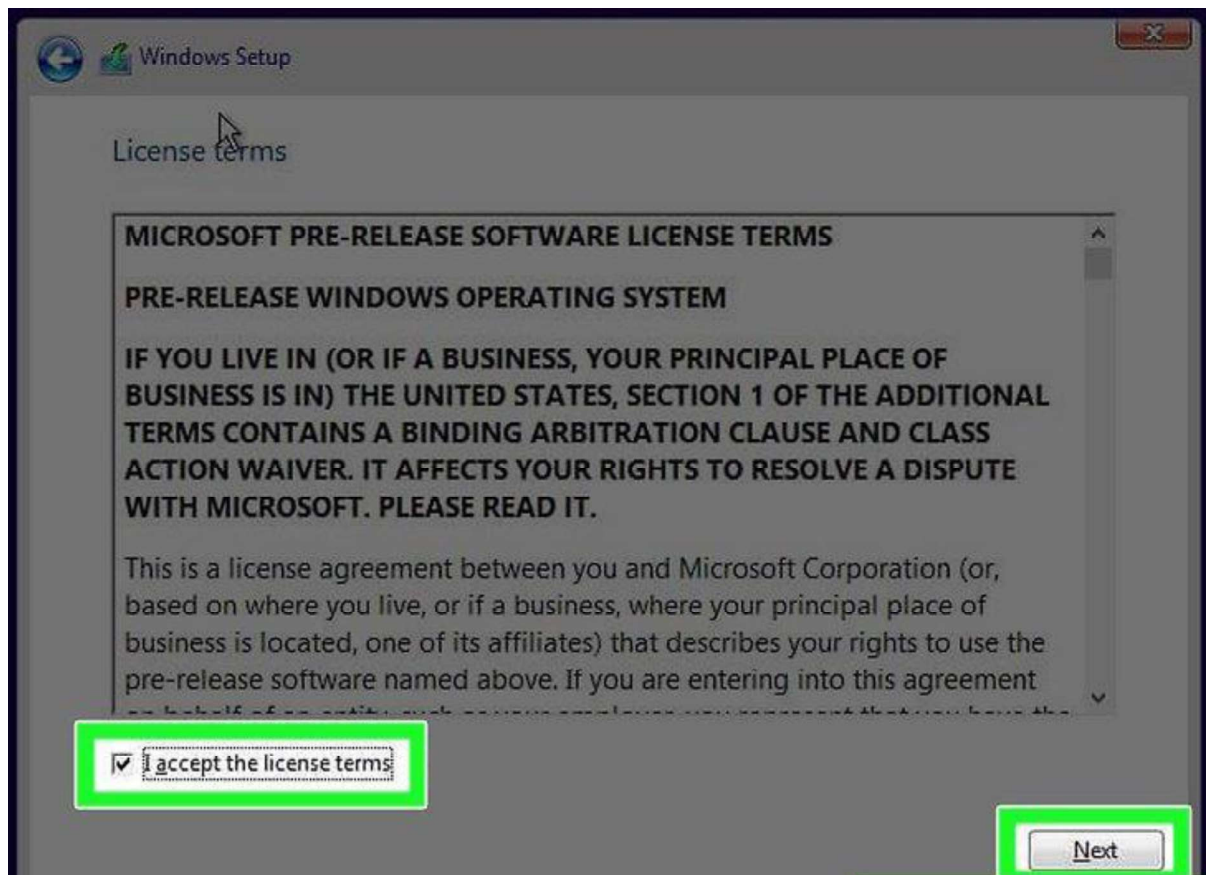


Fig. 2.7 Windows Installation: Accept Terms

In the next window, it will ask: "Which type of installation do you want?". Upgrade option installs Windows 10 while preserving your files, apps, and settings. Wait for Windows 10 to install. This process may take anywhere from half an hour to several hours, depending on your computer's previous operating system and processing speed.

Installing Drivers

Driver updates for Windows 10, along with many devices, such as network adapters, monitors, printers, and video cards, are automatically downloaded and installed through Windows Update. You probably already have the most recent drivers, but if you'd like to manually update or reinstall a driver, follow the steps given below:

Update the device driver

1. In the search box on the taskbar, enter device manager, then select **Device Manager**.
2. Select a category to see names of devices, then right-click (or press and hold) the one you'd like to update. See Figure 2.8.
3. Select **Search automatically for updated driver software**.
4. Select **Update Driver**.
5. If Windows doesn't find a new driver, you can try looking for one on the device manufacturer's website and follow their instructions.

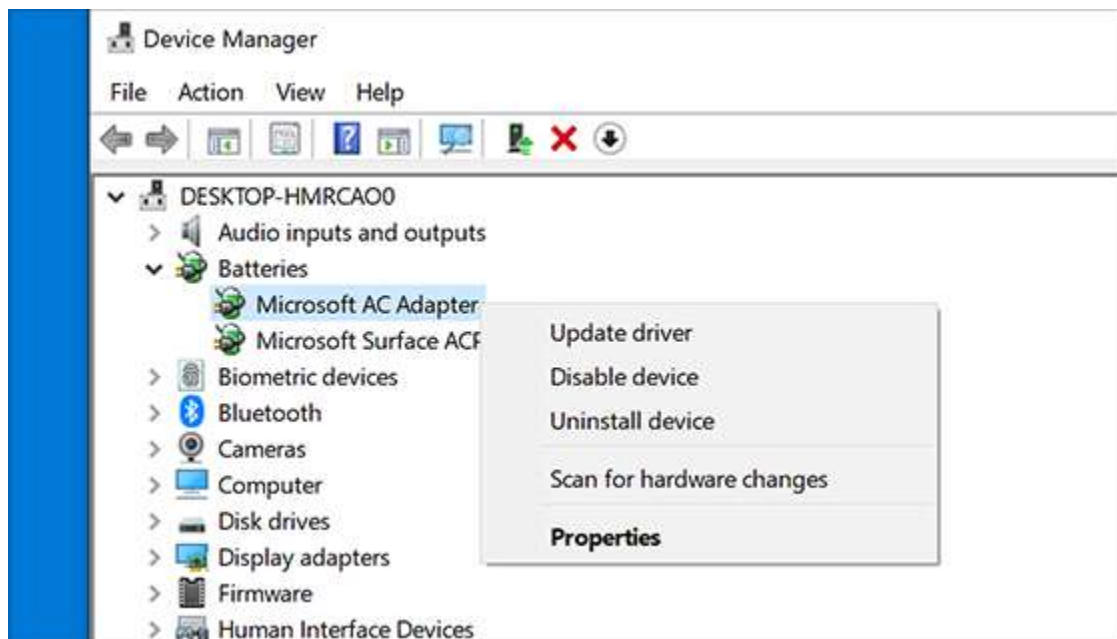


Fig. 2.8 Install Drivers in Windows

2.6 Install New OS in VMware Workstation

Virtual Machine

A Virtual Machine (VM) is a software resource that uses software instead of a physical computer to run programs and deploy apps. One or more virtual “guest” machines run on a physical “host” machine. Each virtual machine runs its own operating system and

functions separately from the other VMs, even when they are all running on the same host.

Follow the following steps to install operating systems on a virtual machine (VMware: commonly used Virtual Machine).

Step 1: Install VMware workstation. Go to **File > New Virtual Machine** (see Figure 2.9).

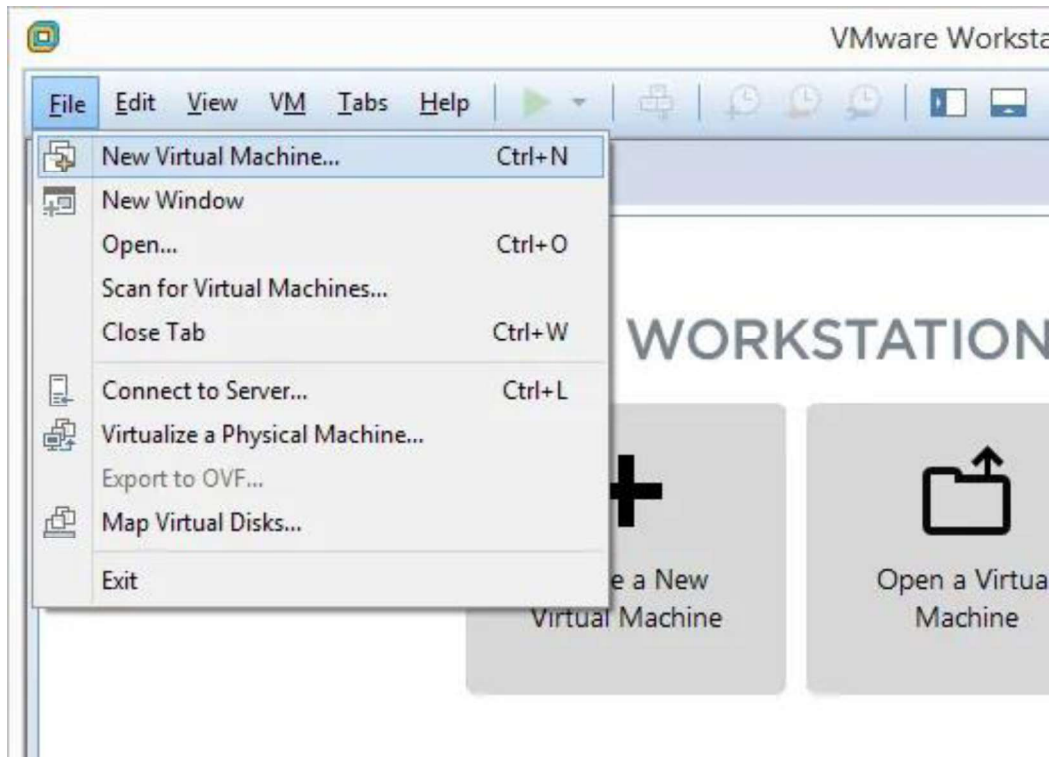


Fig. 2.9 VMware: Select New Virtual Machine

Step 2: Select *Typical (recommended)*, and then press *Next*(see Figure 2.10).



Fig. 2.10 VMware: Select Recommended Settings

Step 3: Pick how to install the operating system. Select **Installer disc** if the operating system is in the disc drive. Otherwise, choose **Installer disc image file (iso)** if you have a file that contains the OS, like a Windows 10 ISO or an ISO for macOS. Refer to Figure 2.11.

Step 4: Choose precisely where the OS files are located.

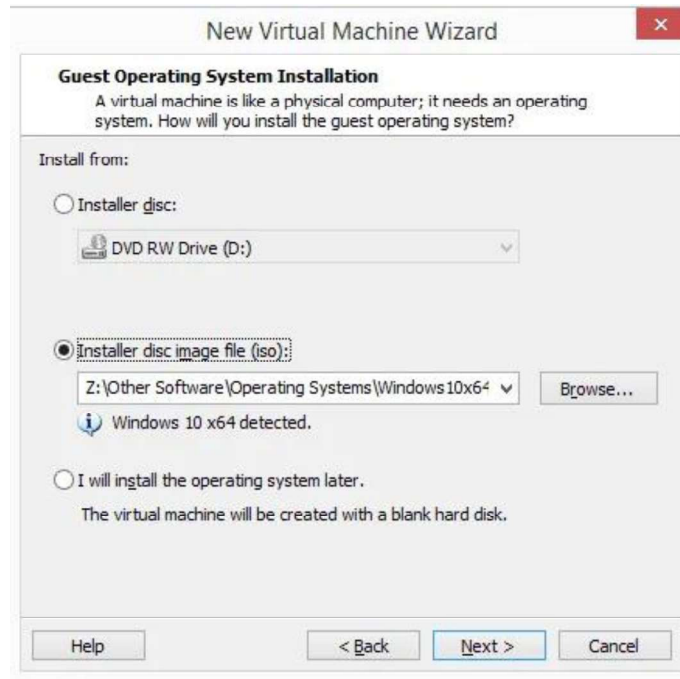


Fig. 2.11 VMware: Select Image File

Step 5: Select storage for this virtual machine, and then press **Next**. Refer to Figure 2.12.

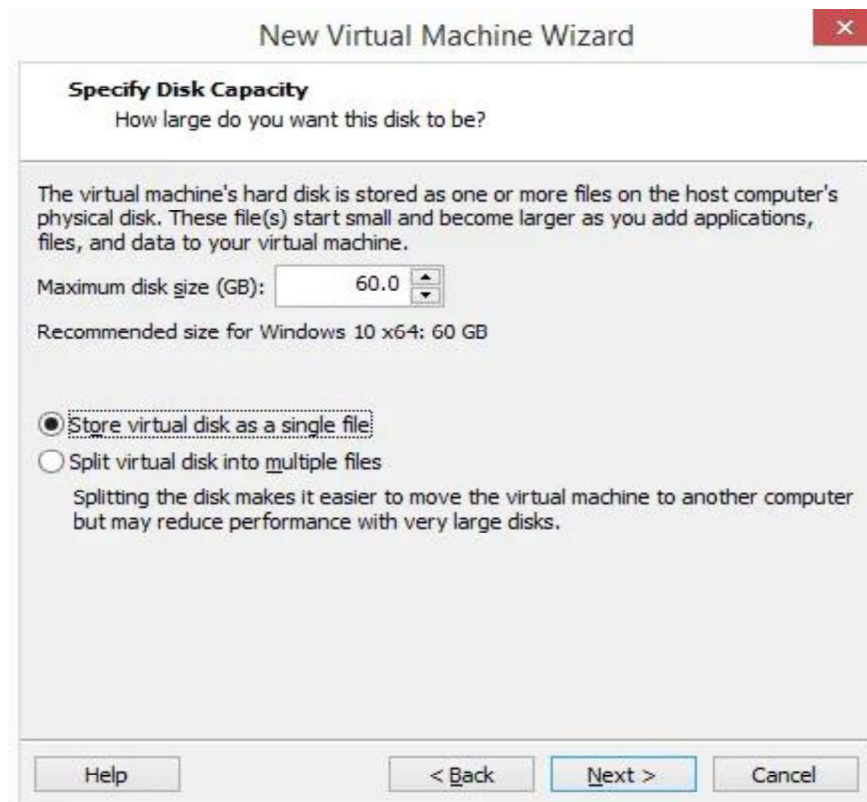


Fig. 2.11 VMware: Select Storage for Virtual Machine

Step 6: Select **Customize Hardware** and make any necessary changes. You can change details about the memory, processors, disc drive, network adapter, USB controller, sound card, printer, and display.

Network Adapter is an important component here that you might end up having to adjust later if your virtual machine doesn't have internet access. There are multiple options, and some may or may not work properly depending on how your host computer is set up and how the guest OS (the VM) works. You can skip this for now to have the default option chosen for you.

Step 7: Choose **Close** to exit the *Hardware* screen, and then press **Finish**. VMware Workstation will create the virtual disk you specified in Step 5 and then turn on the virtual machine automatically. This process might take a while, but you can watch the progress bar for an estimation of when it will finish.

Step 8: Follow the directions for your specific operating system install to add the OS to VMware Workstation.

2.7 Operating Systems Usage

User Interface and commands are used to operate any operating system. We enlist some common ways of using Windows and Linux:

2.7.1 Linux Usage:

1. Cp command

The cp command stands for **copy** is used to **copy files and directories** in Linux System. The syntax for cp command is: cp <Source><Destination>. We can simply use the cp command along with the **source** and **destination** as shown in Figure 2.12.



```
linux-learning@NAG1-LHP-N06778:~$ ls
Desktop    Downloads      file1.txt  Music      Public      Videos
Documents  examples.desktop file2.txt  Pictures   Templates
linux-learning@NAG1-LHP-N06778:~$ cp file1.txt file2.txt
```

Fig. 2.12 Use of cp Command

2. Mv command

The mv command stands for **move** is used to **move files and directories** in Linux System. The syntax for mv command is: mv <source><destination>. We can simply use the mv command along with the **source** and **destination** as shown in Figure 2.13.



```
linux-learning@NAG1-LHP-N06778:~$ ls
Desktop    Downloads      file1.txt  logs1  Pictures  Templates
Documents  examples.desktop file2.txt  Music  Public    Videos
linux-learning@NAG1-LHP-N06778:~$ mv file2.txt logs1
linux-learning@NAG1-LHP-N06778:~$ ls . logs1
.:
Desktop    Downloads      file1.txt  Music      Public      Videos
Documents  examples.desktop logs1      Pictures    Templates

logs1:
file2.txt
```

Fig. 2.13 Use of mv Command

3. Mv command for Renaming

The mv command is used for both moving as well as renaming a file/directory. The syntax for mv command for renaming is: mv <current_name><new_name>. To rename a file we just used mv command along with current_name and new_name as shown in Figure 2.14.

A terminal window with a dark gray border and three colored window control buttons (red, yellow, green) in the top-left corner. The terminal text shows a user named 'linux-learning' at a host 'NAG1-LHP-N06778' in their home directory. They run 'ls' and see a list of directories including 'original'. Then they run 'mv original new' and run 'ls' again to see that 'original' has been replaced by 'new'.

```
linux-learning@NAG1-LHP-N06778:~$ ls
Desktop  Downloads      logs1  Music    Pictures  Templates  Videos
Documents examples.desktop logs2  original Public    text_files
linux-learning@NAG1-LHP-N06778:~$ mv original new
linux-learning@NAG1-LHP-N06778:~$ ls
Desktop  Downloads      logs1  Music    Pictures  Templates  Videos
Documents examples.desktop logs2  new      Public    text_files
```

Fig. 2.13 Renaming using mv Command

4. Search Files and Folders

find command expressions look like this:

find command options *starting/path* expression

The options attribute controls the behaviour and optimization method of the *find* process. The *starting/path* attribute defines the top-level directory where the *find* command in Linux begins the filtering process. The *expression* attribute controls the assessments that scour the directory tree to create output.

Let's break down a Linux *find* command where we don't just want Linux *find* file by name:

```
find -O3 -L /var/www/ -name "*.html"
```

It enables the top-level optimization (-O3) and permits *find* to follow symbolic links (-L). The *find* command in Linux searches through the whole directory hierarchy under /var/www/ for files that have .html on the end.

Basic Examples

1. `find . -name thisfile.txt`

If you need to know how to find a file in Linux called *thisfile.txt*, it will look for it in current and sub-directories.

2. `find /home -name *.jpg`

Look for all .jpg files in the /home and directories below it.

3. `find . -type f -empty`

Look for an empty file inside the current directory.

4. `find /home -user randomperson -mtime 6 -iname ".db"`

Look for all .db files (ignoring text case) that have been changed in the preceding 6 days by a user called *randomperson*.

2.7.2 Windows Usage

1. Personalization

Windows 10 makes it easy to customize the look and feel of your desktop. To access the **Personalization settings**, right-click anywhere on the desktop, then select **Personalize** from the drop-down menu as shown in Figure 2.14. The Personalization settings will appear.

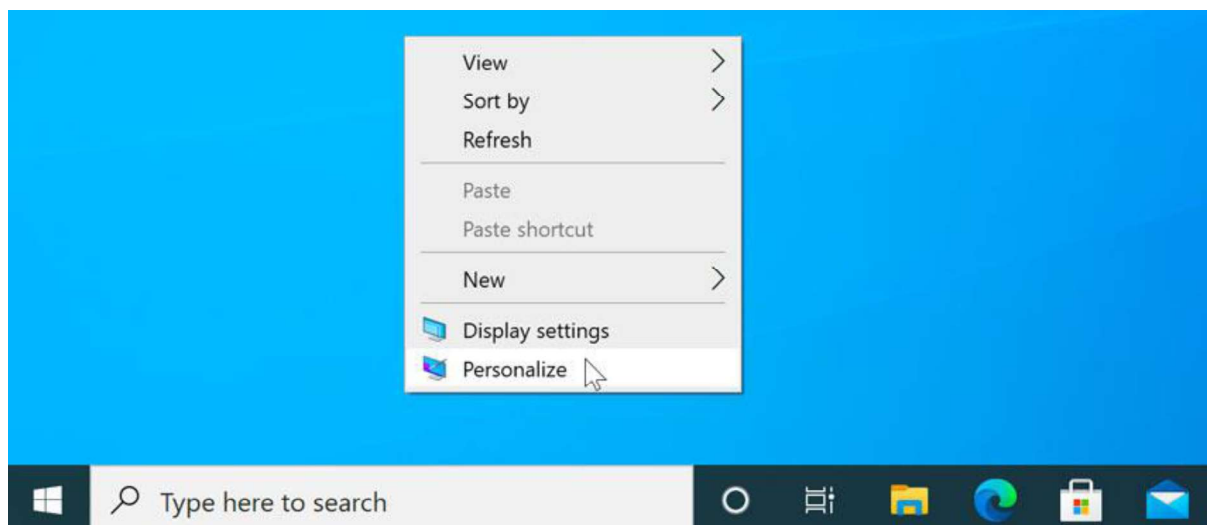


Fig. 2.14 Personalize Settings

2. To change the font size

If you have difficulty seeing the text on your computer, you can increase the **font size**. Increasing the font size will also increase the size of icons and other items on your desktop.

1. Open the **Settings** app, then select **System**.
2. The **Display** options will appear as shown in Figure 2.15. Use the drop-down arrow to scale up or down the size. Note that a larger size may interfere with the way some items appear on the screen.

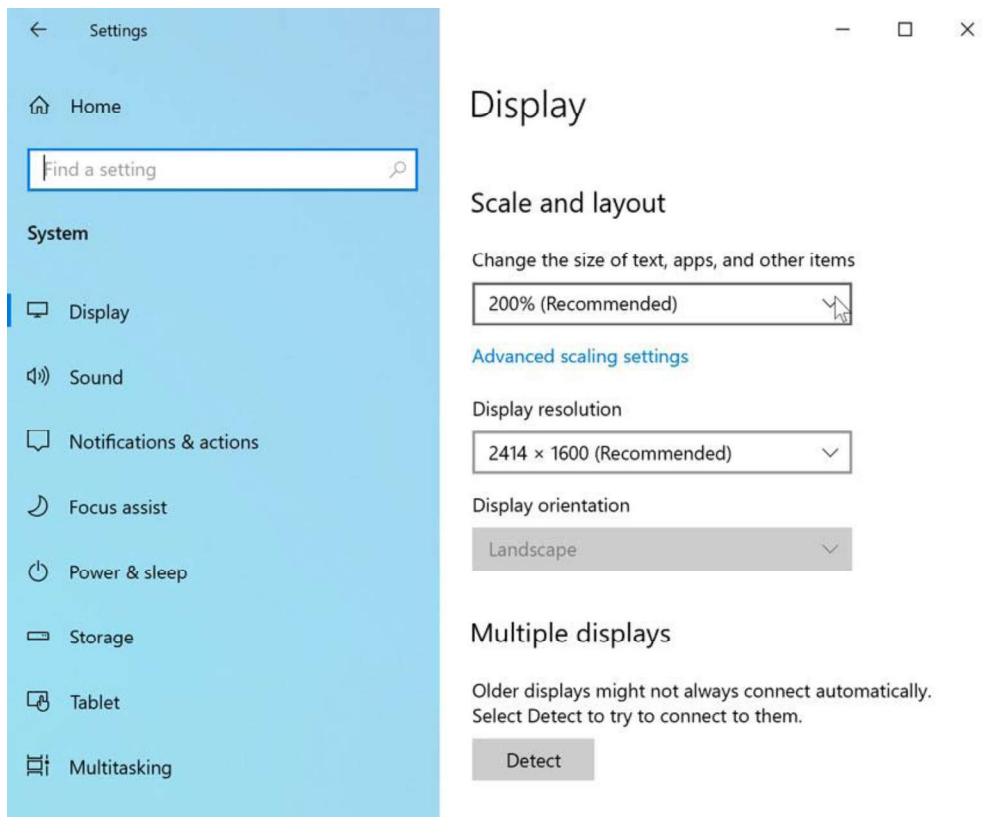


Fig. 2.15 Display Settings

3. Once you've made your selection, the changes will take effect.
4. When you're done, click **Finish**. The ClearType settings will be applied.

3. Windows File Extensions

Windows file names have two parts; the file's name, then a period followed by the extension (suffix). The extension is a three- or four-letter abbreviation that signifies the file type. For example, in **letter.docx** the filename is **letter** and the extension is **.docx**. Extensions are important because they tell your computer what icon to use for the file, and what application can open the file. For example, the **.doc** extension tells your computer that the file is a Microsoft Word file.

4. Unhide Files and Folders

1. Click on the start menu and type "**Show hidden files and folders**," then press the enter key or select that option from the list of results as shown in Figure 2.16.

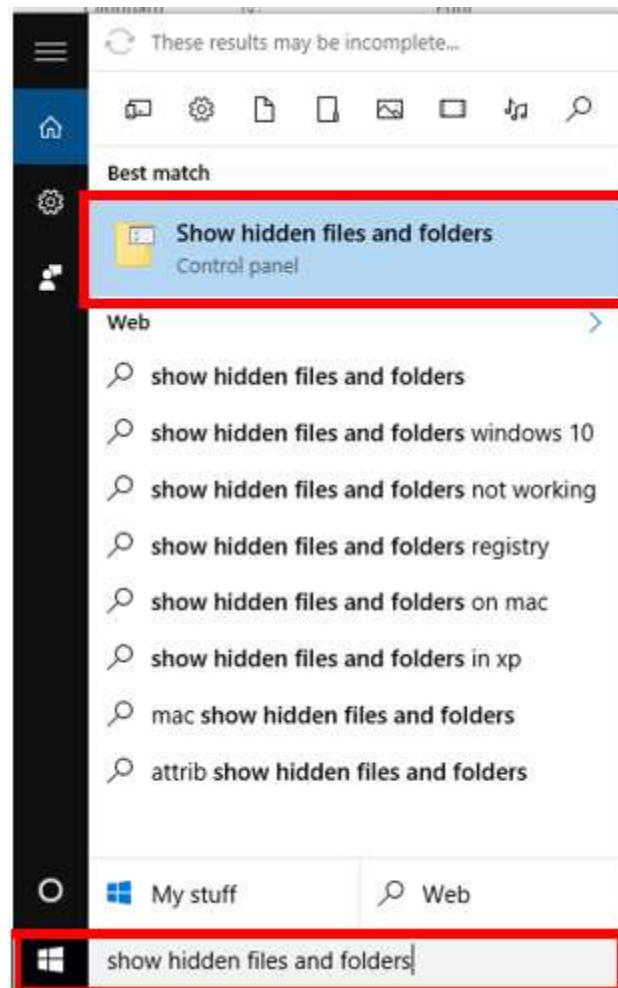


Fig. 2.16 Windows: Show Hidden Files

2. Under **Advanced settings**, select **Show hidden files, folders, and drives**, and then select **OK**.

5. Hide a File or Folder on Windows

To hide a file or folder on Windows, open a Windows Explorer or File Explorer window and locate the file or folder you want to hide. Right-click it and select Properties. Enable the Hidden checkbox on the General pane of the Properties window as shown in Figure 2.17. Click OK or Apply and your file or folder will be hidden.

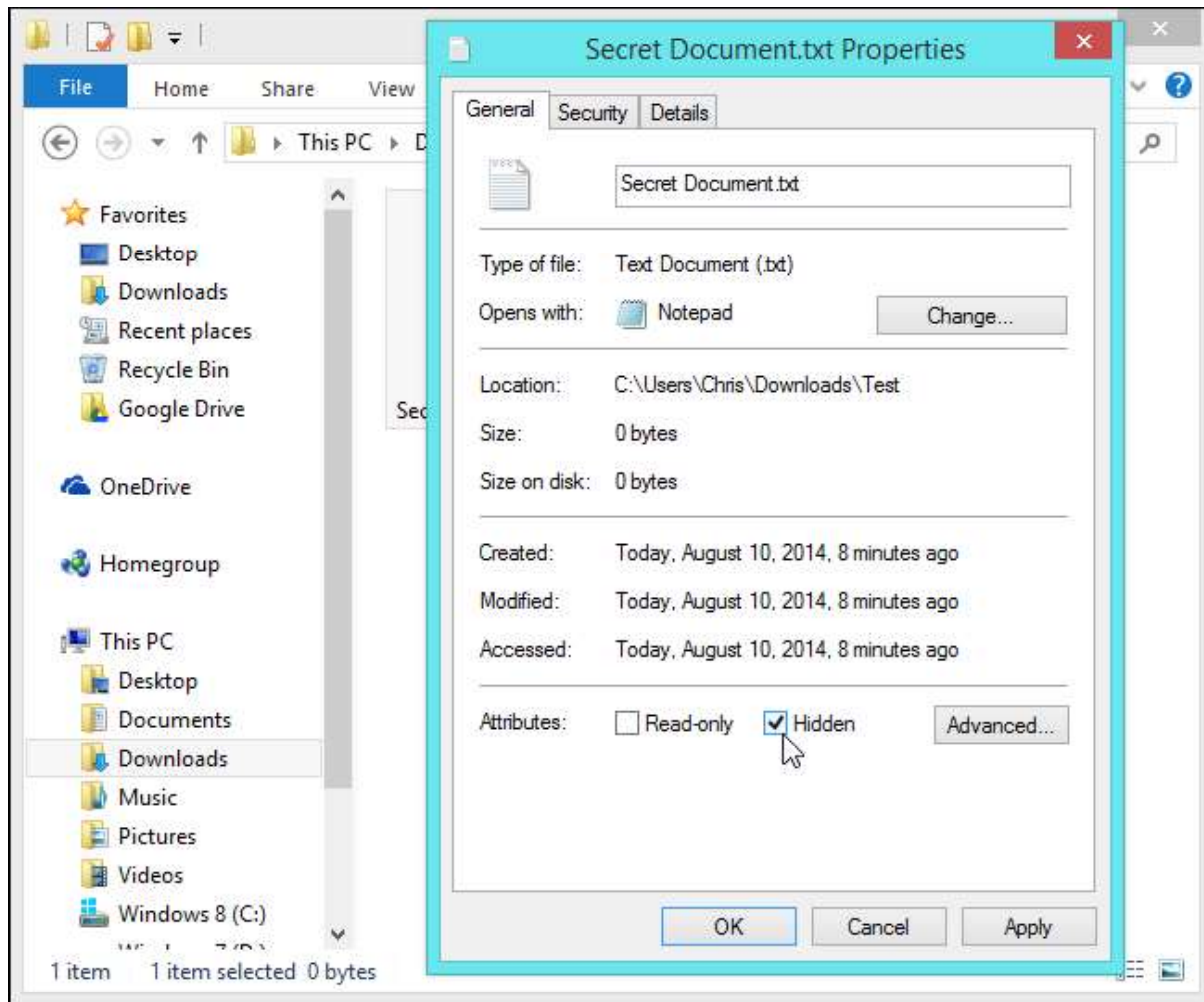


Fig. 2.17 Windows: Hide Files

6. Device Manager

Device Manager is defined as a Control Panel that is an extension of the Microsoft Management Console. It provides users an organized, central view of the Windows-recognized hardware attached to a computer. This includes devices like keyboards, hard disk drives, USB devices etc. It sorts out all hardware, listing them on the basis of various criteria and when a device malfunctions or stops working altogether, it immediately notifies the user. Thus, it can be used to control devices, enable or disable them, configure them and identify conflicts between devices etc.

7.Task Manager

The Windows Task Manager is a powerful tool packed with useful information, from your system's overall resource usage to detailed statistics about each process.

Launching the Task Manager

Windows offers many ways to launch the Task Manager. Press Ctrl+Shift+Esc to open the Task Manager with a keyboard shortcut or right-click the Windows taskbar and select “Task Manager.” You can also press Ctrl+Alt+Delete and then click “Task Manager” on the screen that appears or find the Task Manager shortcut in the Start menu.

The first time you launch the Task Manager, you'll see a small, simple window. This window lists the visible applications running on your desktop, excluding background applications. You can select an application here and click “End Task” to close it. This is useful if an application isn't responding—in other words, if it's frozen—and you can't close it the usual way. Figure 2.18 shows a view of a task manager.

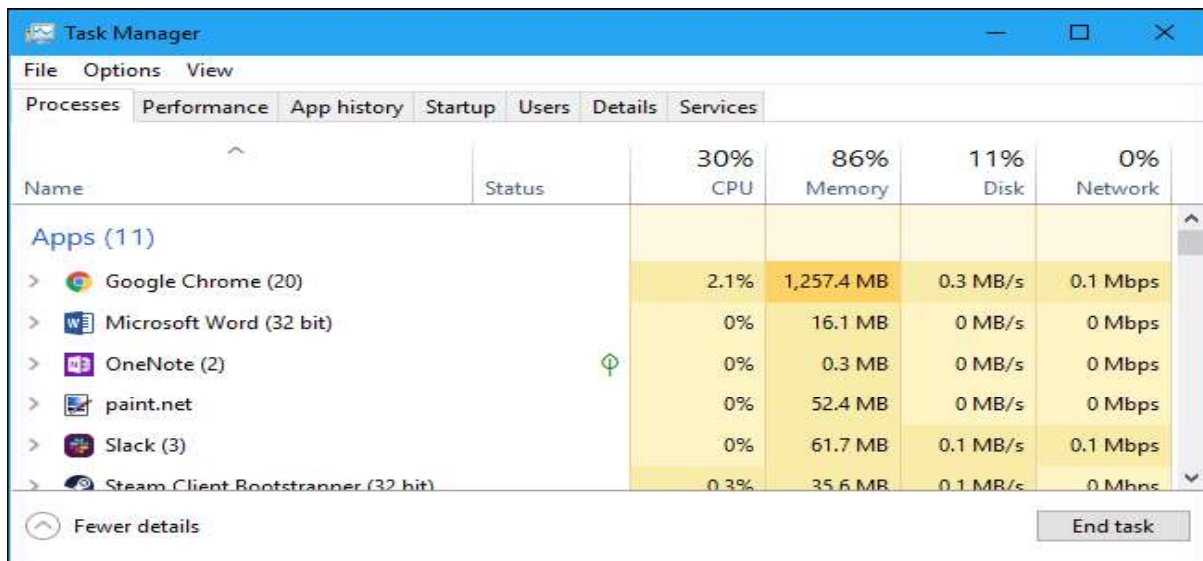


Fig. 2.18 Windows' Task Manager

8. Using the Windows' Command Prompt

Command Prompt is a command line interface for the Windows operating systems. Follow the following steps to open command prompt for Windows:

1. Click Start.
2. In the *Search* or *Run* line, type **cmd** (short for command), and press Enter.

Command prompt in Windows can be used to list contents of a certain directory, move into a directory, run application programs etc. A view of command prompt is given in Figure 2.19.

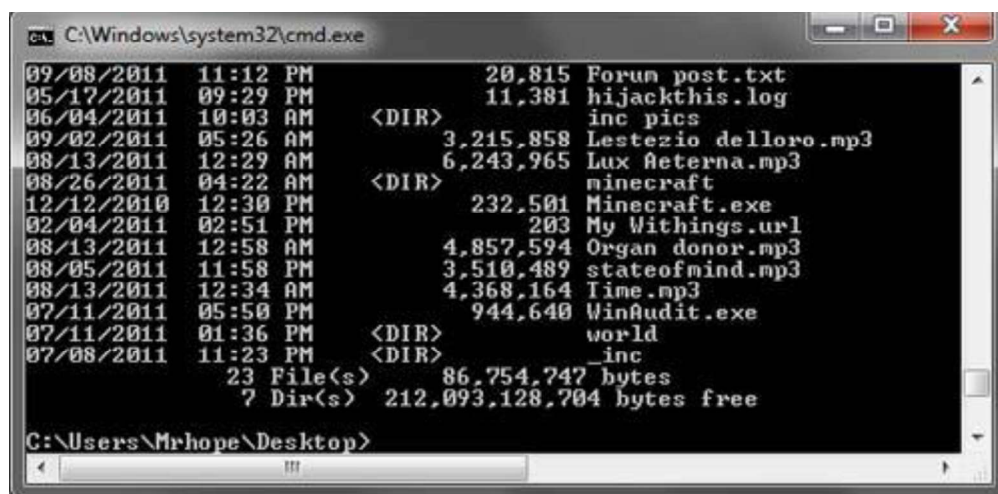


Fig. 2.19 Windows' Command Prompt

Installing an Application Software on Windows

To install programs from a CD or DVD:

1. Insert the program disc into your computer's disc drive or tray, label side up (or, if your computer has a vertical disc slot instead, insert the disc with the label side facing left). The AutoPlay dialog box appears as shown in Figure 2.20.



Fig. 2.20 Windows' AutoPlay Dialog Box

2. Click the option to run Install or Setup. User Account Control may ask if you really want to run this program. If nothing happens when you insert the disc, choose Start→Computer to open the computer window; double-click the icon for your DVD or CD drive; and then double-click a program named Setup or Install.
3. If the installer offers a language selection, choose yours; likewise, agree to the end-user license agreement, if requested. Many installers require you to accept an end-user license agreement (EULA).
4. If the installer offers Express or Custom installation options, choose the Express option to let the installer set up the program without further input from you. The

Custom option or Advanced Settings allows you to specify where to install the program and, perhaps, which parts of the program to install. Some installers provide other options to install documentation and other programs or to register the program.

5. As the installer program continues to run and display dialog boxes, click the Next or Continue button. You proceed to the next dialog box each time you click.
6. Click the Finish or Close button in the last step of the installer program to complete the process.

In a few cases, the setup program may ask you to restart Windows 7. In this case, you don't have to restart immediately, but you won't be able to use the new program until you do restart. The new program may appear automatically at the bottom of the first screen of the Start menu. Look under All Programs as well. Newly installed programs are highlighted in colour. Some installers add a program icon to the desktop.

Many programs try to connect to the Internet for updates during installation or when you run the installed program. The first time you run a program, you may be asked if you want to register the program or configure some aspect of the program. Go with default (assumed) responses, if you're not sure.

2.8 Update vs Upgrade

Updating involves making changes to an app or an operating system in such a way that it doesn't affect its core structure. So, most of the frequent changes made to your computer like bug fixes, security patches, adding support for drivers and newer hardware, etc. can be termed as an Update. An update is often small in size, and it might take a couple of minutes to perform one. The word 'small' is relative. For instance, an update meant for a single app can range from a few kilobytes to a couple of megabytes in size, while an update for an operating system can go up to a few hundred megabytes.

When a set of changes made to a software are significant and substantial enough, it is called an Upgrade. That's why making a switch from Ubuntu 16.04 to Ubuntu 17.04

would be called an upgrade, not update. An upgrade mostly includes important changes to the GUI and a variety of new features and options which are not in the existing version of a software or operating system. And as you might have guessed, its size can go up to several gigabytes.

Updating Windows

To manage update options and see available updates, select Check for Windows updates. Or select the Start button, and then go to Settings > Update & Security > Windows Update.

Upgrade from Windows 7 to Windows 10

Users can either do an in-place upgrade on same computer or a new clean install on a new PC. In both methods simply use the Windows 7 Serial key when asked and it will work perfectly. Make sure you have enough free disk space before upgrading. Check the minimum free disk space required for Windows 10 upgrade.

In-place Upgrade

In place upgrade helps to upgrade your existing Windows7 or Windows8 system to the latest windows10 by keeping all your programs and data migrated to the latest version. However, we recommend you to take the backup of your data before starting the procedure.

Step 1: Download & run media creation tool.

Step 2: Select the option ‘**upgrade this PC**’. When you run the media creation tool it will give you two options as shown in Figure 2.21. Select the first one “Upgrade this PC now”.

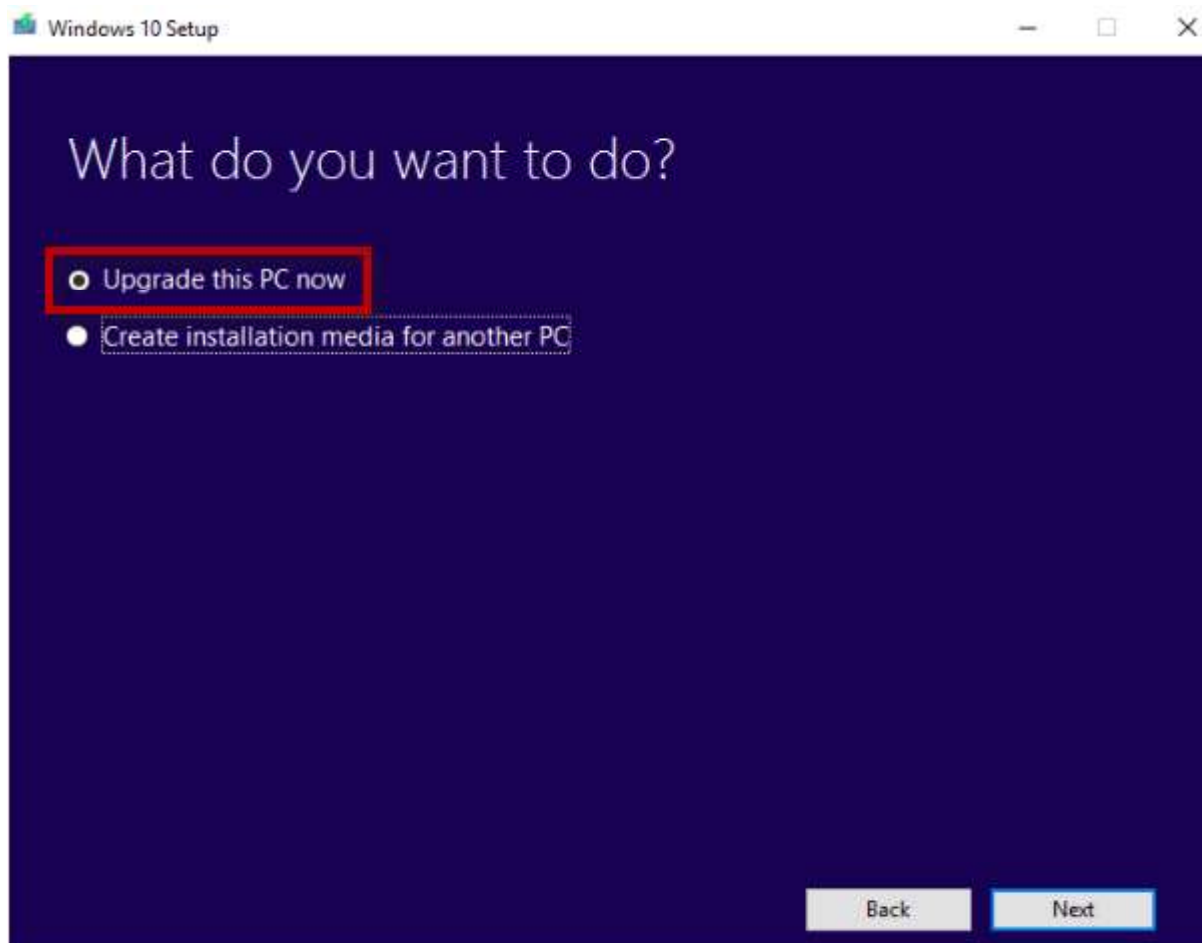


Fig. 2.21 Windows' Media Creation Tool: Upgrade Options

Step 3: Wait while it downloads the required files and then click 'Install' button.

It may take some time in downloading Windows 10 files from the internet. So, it depends on your internet speed that how much time will it take. If you are using a pirated or cracked version then it may ask you to enter the product key. Otherwise, it will not ask for any product / license key and upgrade process will go smoothly. After completion of download, click on the '**Install**' button.

Step 4: Select 'Keep the personal files and apps' option as shown in Figure 2.22. After clicking on the **install** button, it will ask for what to keep. You can choose data only, data & apps or nothing. By default, it will select the '**data & apps**' option.

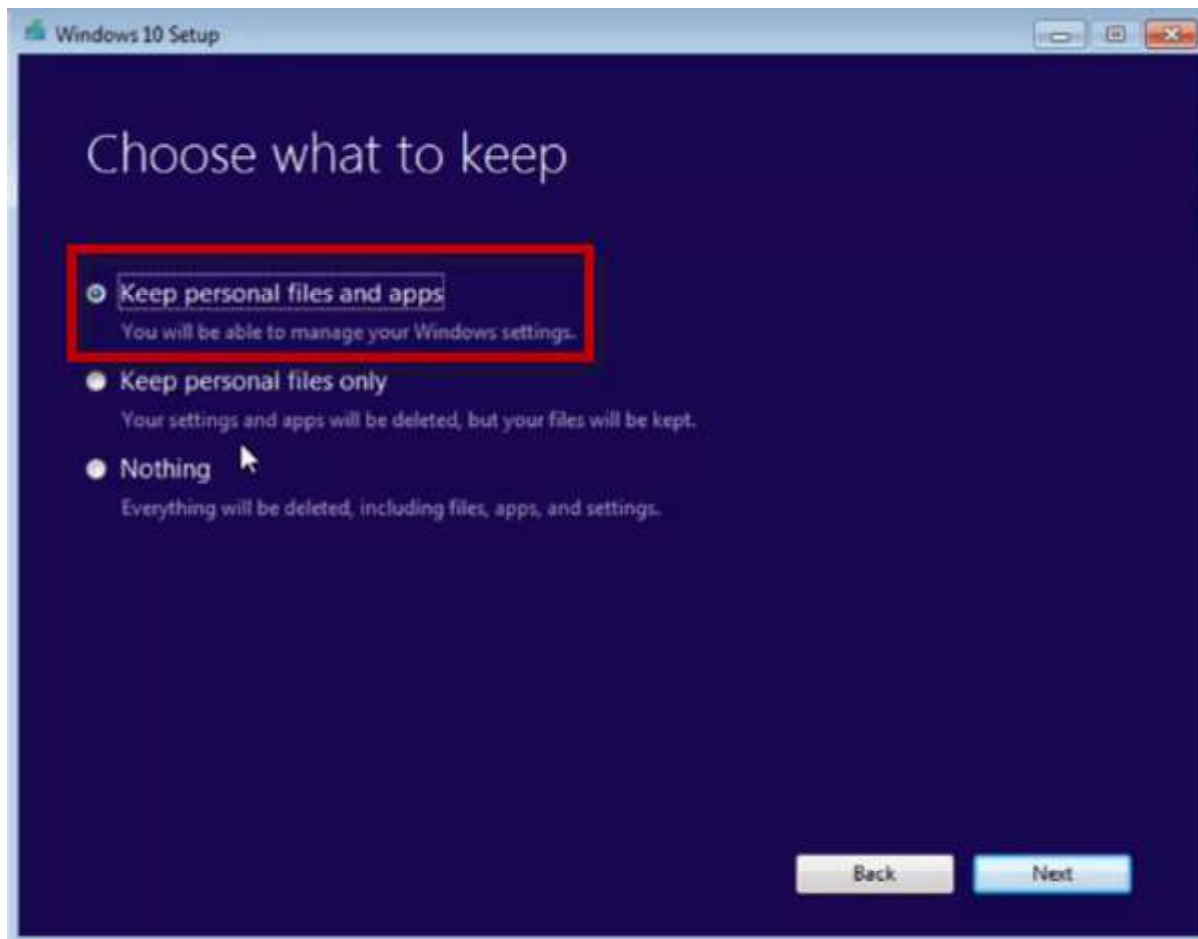


Fig. 2.21 Windows' Media Creation Tool: Backup Options

Step 5: Wait while upgrade process finishes. Click on the **next** button and it will start the upgrade process and may restart the system several times.

Step 6: If everything goes smooth you are welcome by Welcome screen of Windows 10. It will ask you to set the privacy settings first. Set the privacy settings and you are ready to go.

Key Points

- A software is a set of instructions or programs instructing a computer to do specific tasks.
- System software is an interface between application software and system.
- Application software is the type of software which runs as per user request. It runs on a platform which is provided by the system software.
- An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs.
- A firmware is a tangible electronic component with embedded software instructions, such as a BIOS.
- A Virtual Machine (VM) is a compute resource that uses software instead of a physical computer to run programs and deploy apps.
- The cp command stands for copy is used to copy files and directories in Linux System.
- The mv command stands for **move** is used to **move files and directories** in Linux System.
- It provides users an organized, central view of the Windows-recognized hardware attached to a computer.
- Updating involves making changes to an app or an operating system in such a way that it doesn't affect its core structure.
- When a set of changes made to a software are significant and substantial enough, it is called an Upgrade.

Exercise

Select the most appropriate option.

1. Windows is a
 - a. Operating Systems
 - b. Firmware
 - c. Application Software
 - d. Both a and b
2. VMware is a
 - a. Operating Systems
 - b. Firmware
 - c. Application Software
 - d. Both a and b
3. Which of the following OS is open source?
 - a. Linux
 - b. Windows
 - c. MSDOS
 - d. None of these
4. cp command is used to
 - a. move files
 - b. copy files
 - c. delete files
 - d. none of these
5. mv command is used to
 - a. move files
 - b. copy files
 - c. delete files
 - d. none of these

Answer the following questions.

1. Define Software.
2. Differentiate between applications software and system software?
3. Describe functionalities of an operating system.
4. Define Firmware.
5. Define Virtual Machine.
6. Define Task Manager.
7. Describe command prompt and its usage.
8. What is the purpose of Device Manager in Windows?
9. Differentiate Update and Upgrade.

Practical Tasks

1. Install Windows 10 on the given PC.
2. Replace Windows Installation with the Ubuntu Installation.
3. Install Ubuntu on VMware Workstation.
4. Use Command Prompt to copy the contents of a file into another file.
5. Uninstall an application program from Windows using Control Panel.
6. Upgrade Windows 7 to Windows 10 and make sure to keep all programs and data in the Windows 10.